EPSR-test-ex78 sdf orientation test (DMSO), sdf water vs S=O (distinct), mapgrentropy

Program(s): sdf, mapgrentropy | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the orientation test of `sdf` [1] for identical four-atom groups and a distinct two-atom second group, together with accumulation from a missing file and `mapgrentropy` on `sdf` coefficients.

Code exercised

Fortran (original, fortran_src/):

- `sdf_rev2.f`: `run_sdf` (keywords), `sdf` (atom-group search by type and distance, pair loop, averaged groups)
- `sharm_routines_rev1.f`: `get_l_and_n_values`, `form_cgcoeffs`, `form_eulerangles`, `form_dlmn`, `form_sum_prod_dlmn_cgcoeff`
- `auxiliary_routines.f`: `dlm`, `cgcoef`, `rotateeuler`; `read_and_save_files.f`: `outgofzs_sharm` (`.h0n` files, averaging); `process_symbols.f`: `check_symbol`
- `mapgrentropy.f`: `run_mapgrentropy` (entropy sums); `auxiliary_routines.f`: `initran1`, `saveran1`; `plotsharm_routines.f`: `smoosharm`, `sumoverm2`

Rust (port, rust/epsr-core/src/):

- `tools/sdf.rs`: `run`, `sdf`, `check_symbol_extra`
- `tools/sharm_routines.rs`: `Sharm` methods with the Fortran names (incl. `outgofzs_sharm`), `dlm`, `cgcoef`, `form_eulerangles`, `rotateeuler`
- `tools/mapgrentropy.rs`: `run`, `initran1`; `rng.rs`: `Ran1::ran1`; `sharm_routines.rs`: `Coeffs::read`, `Coeffs::angular`

Physical system

The EPSR-test configuration `200DMSO400Water.ato`: 200 DMSO (atoms S, C, C, O, 6 H) and 400 water molecules (O1, H1, H1; SPC/E parameters [2]), 3200 atoms at 300 K in a cubic box of side 33.12 Å, a model of the DMSO-water mixtures of Soper and Luzar [3]. `sdf` computes the same spherical-harmonic coefficients [4,5] as `sharm`, but its "molecules" are groups of atoms found by atom types and pair distances (first atom = centre). `dmsotest`: groups S O C C with all six pair distances (±15 %), `test 15 25 ...`, 97 coefficients. `watso`: central water (O1 H1 H1) and the S=O pair of DMSO (1.53 Å ±10 %) as second group, l2 = 0-1, 27 coefficients. `mapgrentropy` adds to the ideal-gas entropy the two-body excess term, an integral of $r^2 \cdot (g - 1 - g \ln g)$ [6], for the radial, spatial and full g, sampling random orientations with `ran1` [7].

Inputs

Input files written by hand from the keyword tables (`setup_symbols_*`), one `keyword value` per line, ended by `q`; `200DMSO400Water.ato` copied from EPSR-test. `dmsotest.SDF.dat` (`averagemol n`); `watso.SDF.dat` with `nsumt 5`; `watso.mapgrentropy.dat`: 80 trials, smoothing `2.0 1 1 1`.

How the accuracy was verified

`commands.txt` (3 program calls, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. A run-time FORMAT in the original source (`sharm_rev2.f`/`sdf_rev2.f`, and `sharm_diffuse` for `epsr`) lacks a comma after the Hollerith `1h(` and is rejected by current gfortran, so the reference build inserts that comma (originals kept as `*.orig`); the port behaves like the patched program. Result of the final regression run of the complete port: every file identical in all 2 passes.
Two passes. `watso`: pass 1 finds no previous file, pass 2 skips the calculation (final `nsumt 7`); the second molecule is written from the averaged second-group coordinates. The orientation test leaves `nsumt 0` and is recomputed in pass 2.
Reference build: `sharm`/`sdf` were rebuilt with one comma inserted in the run-time title format (`'(1h(,'//il1fmt`), since gfortran 15.2 otherwise stops with "Missing comma between descriptors" before writing the coefficients; all other programs are the unmodified reference binaries.

Notes

Fortran quirk reproduced: `run_sdf` does not include `sharm_sdf.inc`, so the test angles it reads are local variables and the angles actually used by `sdf` are all zero.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[3] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[4] A. K. Soper, "Orientational correlation function for molecular liquids: The case of liquid water", J. Chem. Phys. 101, 6888-6901 (1994).
[5] C. G. Gray and K. E. Gubbins, Theory of Molecular Fluids, Vol. 1: Fundamentals, Clarendon Press, Oxford (1984).
[6] A. Baranyai and D. J. Evans, "Direct entropy calculation from computer simulation of liquids", Phys. Rev. A 40, 3817-3822 (1989).
[7] S. K. Park and K. W. Miller, "Random number generators: good ones are hard to find", Commun. ACM 31, 1192-1201 (1988).
[8] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.